

Hayeon Bang

hayeonbang@kaist.ac.kr
website | github | linkedin

I Researcher, Composer

Research interest: Educational Music AI, Human-AI Musical Interaction

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Ph.D. student in Graduate School of Culture Technology Music and Audio Computing (MAC) Lab Advisor: Prof. Juhan Nam	Feb 2025 - present
Korea Advanced Institute of Science and Technology (KAIST) M.S. in Graduate School of Culture Technology Music and Audio Computing (MAC) Lab Advisor: Prof. Juhan Nam	Mar 2022 - Feb 2025
Yonsei University Bachelor's Degree in Music Composition	Feb 2016 - Feb 2021

PUBLICATIONS

D3PIA: A Discrete Denoising Diffusion Model for Piano Accompaniment Generation Eunjin Choi, Hounsung Kim, Hayeon Bang, Taegyun Kwon and Juhan Nam IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2026	
PianoBind: A Multimodal Joint Embedding Model for Pop-piano Music Hayeon Bang, Eunjin Choi, Seungheon Doh and Juhan Nam Proceedings of the International Society of Music Information Retrieval (ISMIR), 2025	
Dialogue in Resonance: an Interactive Music Piece for the Piano and Real-time Automatic Transcription Hayeon Bang, Taegyun Kwon and Juhan Nam Proceedings of the International Computer Music Conference (ICMC), 2025	
PIAST: A Multimodal Piano Dataset with Audio, Symbolic and Text Hayeon Bang, Eunjin Choi, Megan Finch, Seungheon Doh, Seolhee Lee, Gyeong-hoon Lee and Juhan Nam Proceedings of the 3rd Workshop on NLP for Music and Audio (NLP4MusA), 2024	

PROJECTS

X-Space: Duo Performance for Piano AI Piano KAIST X Daejeon Arts Center (2024) [link] Composer, Music Director	
EasyJam: AI-based Jam Program for Developmental Disabilities Selected for the 2024 Gyeonggi-do Integrated Support Program for Artists with Disabilities, with Mirae-therapy center (2024) Program Developer	
Four Seasons 2050 - Daejeon (2023) [link] Project leader, Composer	

AWARDS & HONORS

Seong-bu Kim Creativity Award (2025) KAIST

High Honor Awards: Academic Achievement for highest grade students, Academic Achievement for high grade students (2020, 2021) | Yonsei University

Scholarship for Human Resource Development (2021) | Mapo-gu

Scholarship for Exchange Program (2019) | Mirae Asset Park Hyeon Joo Foundation
Representative awardee

SELECTED RESEARCH PROJECTS

AI Agent Technology for Interactive Cultural Creation Based on Artists' Individual Characteristics
Korea Creative Content Agency | 2025–present

AI/Big Data-Based Platform for Process-Oriented Assessment of Middle School Music Performance
Collaboration with Ewha Womans University, Music Education | 2024–2026

Deep Learning Models for Improving Music Tagging Performance
Funded by Kakao Entertainment | 2023–2025

Piano Cover Generation for Pop Music with Text-Based Style Control
Funded by NCSoft | 2023–2025

Music-Making in the Post-AI Era
KAIST | 2023

MUSICAL EXPERIENCE

Singer, Piano Arranger at Sumi Jo Talk Concert, KAIST (2024)

Choral Conductor in Woncheon Church 1st Choir (2021 - present)

Actress, Music Director, President in Yonsei University Musical Club ROTHEMS (2016 - 2020)

Orchestra Conducting Education, Sejong Orchestra Conductor Course (2018)

Musical Theater Education, Chungmu Art Center Musical Theater Academy (2018)

TEACHING EXPERIENCE

Teaching Assistant | GCT731 Topics in Music Technology <Music Informatics>, KAIST (2026)

Teaching Assistant | CTP 431 Fundamentals of Computer Music, KAIST (2026)

Instructor | AI for Music Education, Association of Korean Traditional Music Teachers (2026)

Teaching Assistant | GCT 536 Cognitive Science of Music, KAIST (2025)

Instructor | Sonic Visualizer for Music Composers, Yonsei University (2023)

Instructor | Understanding of Music AI, Dangjin Youth Center (2022)

Teaching Assistant | Sound Technology for Multimedia, KAIST (2022)

Instructor | Music Composition, Music Theory, Ear Training, Piano (2017 - 2024)